Jukes–Cantor Distance for the 6502life Noisy Copy Channel

6502life project

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Abstract

We derive a closed-form evolutionary distance formula for cells diverging through the 6502life noisy copy channel, where each bit is independently randomized with probability ε per copy event. This is the binary Jukes–Cantor model. The MLE of evolutionary distance is a closed-form function of Hamming distance.

1 The Noisy Copy Channel

When a BRK noisy-copy fires (operands \$F5–\$F8), the origin cell's $M = 1024$ bytes are copied to a neighbor. Each bit is independently randomized (replaced by a fair coin flip) with probability ε (`pBitNoise`), and faithfully copied with probability $1 - \varepsilon$. The default is $\varepsilon = 1/2048$, giving $\sim$1 bit error per 256-byte page copied.

2 Binary Jukes–Cantor Distance

Proposition 1. *After T copy events on the tree path between two cells, the fraction of differing bits is*

$$p = \tfrac{1}{2}\big(1 - (1 - \varepsilon)^T\big) \tag{1}$$

and the MLE of T given observed bit mismatch fraction $\hat{p}$ is

$$\boxed{\hat{T} = \frac{\log(1 - 2\hat{p})}{\log(1 - \varepsilon)}} \tag{2}$$

Saturation: $p \to 1/2$ as $T \to \infty$.

Proof. Each bit evolves under a 2-state symmetric Markov chain with transition rate $\varepsilon/2$ per copy event. Starting from "same" at $T = 0$: $P(\text{diff}) = \frac{1}{2}(1 - (1 - \varepsilon)^T)$. This is the standard binary Jukes–Cantor model. The MLE of p from n i.i.d. bit comparisons is $\hat{p} = (\text{differing bits})/n$, and the invariance property of MLEs gives the formula for $\hat{T}$. □

2.1 Byte-level distance

Since bits within a byte are independent (no byte-level noise), the probability that two corresponding bytes match is $(1 - p)^8$:

$$P(\text{byte differs}) = 1 - \Big[\frac{1 + (1 - \varepsilon)^T}{2}\Big]^8 \tag{3}$$

Inverting:

$$\hat{T} = \frac{\log\big(2(1 - \hat{p}_{\text{byte}})^{1/8} - 1\big)}{\log(1 - \varepsilon)} \tag{4}$$

2.2 Byte Hamming histogram

The per-byte Hamming distance $h \in \{0, \ldots, 8\}$ follows $\mathrm{Bin}(8, p)$. The sufficient statistic is the sample mean $\bar{h}$, giving $\hat{p} = \bar{h}/8$. The histogram provides no additional information beyond the mean (it is a single-parameter exponential family), but can be used for goodness-of-fit testing against the copy-noise model.

3 Applicability

The distance formula assumes all sequence divergence arises from the noisy copy channel. In practice, cross-contamination (random cells writing into each other's code regions) is often the dominant source of divergence and is not phylogenetically structured. The byte Hamming histogram can diagnose this: under the copy-noise model, $h \sim \mathrm{Bin}(8, p)$; excess weight at $h \approx 4$ relative to $h = 0$ indicates non-copy contamination.